Chapter 9: Correlation

Measuring Relationships Between Variables

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1 Overview

Before You Begin

This chapter introduces correlation — a fundamental tool for understanding relationships between variables. Unlike t-tests, which compare groups, correlation asks whether two continuous variables tend to move together.

This chapter covers:

- Relationships between variables
- Scatterplots and visual patterns
- Pearson's r (for linear relationships)
- Spearman's rho (for ranked data)
- Interpreting correlation strength
- Why correlation does not imply causation

All concepts are illustrated using DataLab examples from your class.

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2 Part 1: Thinking About Relationships

2.1 A New Question

So far, we've focused on comparing groups:

- Do concrete words lead to better memory than abstract words?
- Does mood improve after the DataLab session?

But many research questions ask about **relationships**:

- Do people who sleep more perform better on cognitive tasks?
- Is self-confidence related to actual performance?
- Do extraverts report higher mood?

These questions aren't about group differences. They're about whether two variables **covary** — when one goes up, does the other tend to go up (or down)?

2.2 What Is a Correlation?

Correlation: A statistical measure of the strength and direction of the relationship between two variables.

Correlation answers: When one variable changes, does the other variable tend to change in a predictable way?

2.3 Positive vs Negative Relationships

Positive correlation: As one variable increases, the other tends to increase.

- Sleep hours ↑ → Performance ↑
- Study time ↑ → Exam scores ↑
- Age ↑ → Vocabulary ↑

Negative correlation: As one variable increases, the other tends to decrease.

- Stress ↑ → Sleep quality ↓
- TV time ↑ → Exercise ↓
- Anxiety ↑ → Performance ↓

No correlation: No predictable relationship between variables.

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3 Part 2: Scatterplots — Seeing Relationships

3.1 Why Visualise First?

Before calculating any numbers, you should **always** plot your data. A scatterplot reveals:

- The direction of the relationship (positive, negative, or none)
- The strength of the relationship (tight cluster vs spread out)
- The shape (linear vs curved)
- Any unusual patterns or outliers

3.2 Anatomy of a Scatterplot

Each point represents one participant:

- **X-axis**: One variable (often the predictor or independent variable)
- **Y-axis**: The other variable (often the outcome or dependent variable)

DataLab Example: Self-rating vs actual throwing performance

Participant	Self-rating (1-10)	Actual accuracy (%)
1	7	65
2	4	40
3	8	70
4	5	55

Participant	Self-rating (1-10)	Actual accuracy (%)
5	6	50
…	…	…

On the scatterplot:

- X-axis: Self-rating
- Y-axis: Actual accuracy
- Each dot is one person

3.3 Reading Scatterplots

Pattern Guide:

- **Upward slope** = positive relationship
- **Downward slope** = negative relationship
- **No slope** = no relationship
- **Tight cluster** = strong relationship
- **Wide scatter** = weak relationship

3.4 Example Patterns

```
Strong Positive        Weak Positive         No Relationship
    •                      •  •                  •   •
   • •                   •    •  •            •     •
  •  ••                 •  •   •             •  •    •
 •    •                  •  •                  • •  •
•                       •                    •    •
```

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4 Part 3: Pearson’s r

4.1 The Correlation Coefficient

Pearson’s r is the most common measure of correlation. It quantifies what you see in a scatterplot as a single number.

Pearson’s r: A measure of the **linear** relationship between two continuous variables, ranging from −1 to +1.

4.2 Interpreting r

Value of r	Meaning
r = +1	Perfect positive correlation
r = 0	No linear correlation
r = −1	Perfect negative correlation

The sign tells you the direction; the magnitude tells you the strength.

4.3 Strength Guidelines

Cohen (1988) suggested these benchmarks:

	r
.10	Small
.30	Medium
.50	Large

> ⚠ Context Matters
>
> These are rough guidelines. In some fields (e.g., personality psychology), r = .30 is typical. In others (e.g., physics), r = .90 might be expected.

4.4 What r Actually Measures

Pearson's r measures how well the data points cluster around a straight line.

- **r = 1**: All points fall exactly on an upward-sloping line
- **r = 0**: Points are scattered with no linear pattern
- **r = −1**: All points fall exactly on a downward-sloping line

In practice, you'll rarely see r = 1 or r = −1 in psychology. Real data is messy!

4.5 The Formula (For Reference)

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

You won't need to calculate this by hand — software does it instantly. But the key idea is:

- The numerator measures how much x and y **co-vary** (move together)
- The denominator standardises this to fall between −1 and +1

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5 Part 4: Worked Example — Self-Rating vs Performance

5.1 DataLab Context

In the Frontal Lob task, participants rated how good they thought they'd be at throwing (1-10 scale), then we measured their actual accuracy.

Research question: Do people who rate themselves higher actually perform better?

5.2 The Data

Suppose we have these results from 20 participants:

- Mean self-rating: 5.8
- Mean accuracy: 52%
- Pearson's r = 0.42

5.3 Interpretation

r = 0.42 suggests a **medium positive correlation** between self-rating and actual performance.

In words: People who rated themselves as better throwers tended to be somewhat better — but the relationship isn't perfect. There's substantial individual variation.

5.4 Visualising r = 0.42

```
Accuracy (%)
    |
70  |        •     •
    |      •   •  •
60  |    •  •   •
    |   • •    •
50  |  •  •
    | •
40  |•
    +-----------------
       4    6    8    10
         Self-rating
```

The upward trend is visible, but points don't fall tightly on a line.

5.5 Statistical Significance of r

We can test whether r is significantly different from zero (i.e., whether there's evidence of a relationship in the population).

With n = 20 and r = 0.42:

- t-statistic: 1.95
- df = 18
- p = .067

Since p > .05, we would say: "The correlation was not statistically significant, r(18) = .42, p = .067."

> **i Sample Size Matters**
>
> With larger samples, the same r would be significant. Small samples lack the power to detect moderate correlations.

5.6 Reporting Correlations (APA Format)

> There was a positive correlation between self-rated ability and actual throwing accuracy, although this did not reach statistical significance, $r(18) = .42$, $p = .067$.

Or, if significant:

> Self-rated ability was significantly positively correlated with actual throwing accuracy, $r(58) = .42$, $p = .001$.

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6 Part 5: Spearman's Rho (ρ)

6.1 When to Use Spearman

Pearson's r assumes:

1. Both variables are continuous
2. The relationship is linear

When these assumptions are violated, use **Spearman's rho** (ρ or rs).

Spearman's rho: A correlation based on **ranked** data. It measures monotonic relationships (consistently increasing or decreasing, not necessarily linear).

6.2 Ranked Data

Spearman works by:

1. Converting each variable to ranks (1st, 2nd, 3rd, etc.)
2. Calculating the correlation between the ranks

This makes it robust to outliers and non-linear patterns.

6.3 When to Choose Which

Situation	Use
Both variables are continuous and normally distributed	Pearson's r
One or both variables are ordinal (e.g., Likert scales)	Spearman's ρ
Relationship appears non-linear but monotonic	Spearman's ρ
Data has outliers	Spearman's ρ

6.4 DataLab Example

Rating scales (1-10) are technically ordinal — we can't be sure the difference between 3 and 4 equals the difference between 7 and 8. For the mood ratings in DataLab, Spearman's rho might be more appropriate than Pearson's r.

6.5 Interpretation Is the Same

Spearman's ρ is interpreted identically to Pearson's r:

- Ranges from −1 to +1
- Same strength guidelines (.10 small, .30 medium, .50 large)
- Same APA reporting format

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7 Part 6: Understanding What Correlation Tells You

7.1 What r Squared Tells You

A useful way to understand correlation strength is **r^2** (r squared).

r^2 (coefficient of determination): The proportion of variance in one variable that is "explained by" or "shared with" the other variable.

r	r^2	Interpretation
.10	.01	1% shared variance
.30	.09	9% shared variance
.50	.25	25% shared variance
.70	.49	49% shared variance

Even a "medium" correlation of r = .30 means the two variables share only 9% of their variance. That leaves 91% unexplained!

7.2 The importance of R-squared

! Reality Check

A correlation of r = .42 sounds reasonable. But r^2 = .18 means self-rating explains only 18% of the variance in performance. The other 82% is due to other factors — practice, natural ability, luck, measurement error, etc.

This is why psychology is hard. Human behaviour is influenced by many factors, and any single variable rarely explains more than a small portion.

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8 Part 7: Correlation Does NOT Mean Causation

8.1 The Most Important Lesson

CRITICAL POINT: A correlation between X and Y does **NOT** mean X causes Y. This is possibly the most important thing you'll learn in research methods.

8.2 Why Not?

Three possible explanations for any correlation:

1. **X causes Y**: More sleep → better performance
2. **Y causes X**: Better performance → more sleep (less stress)
3. **Z causes both**: Good health → more sleep AND better performance

Without experimental manipulation, we cannot determine which explanation is correct.

8.3 Classic Examples

Ice cream sales and drowning deaths are correlated.

Does ice cream cause drowning? No! A third variable (hot weather) causes both:

- Hot weather → more ice cream sales
- Hot weather → more swimming → more drowning

Countries with more Nobel Prize winners consume more chocolate.

Does chocolate make you smarter? More likely: - Wealthier countries → more chocolate consumption - Wealthier countries → better education → more Nobel Prizes

8.4 The DataLab Example

Self-rating and throwing performance are correlated ($r = .42$).

Does thinking you're good make you perform better? Maybe — but there are alternatives:

- Past experience → higher self-rating AND better performance
- Natural ability → higher self-rating AND better performance
- Overconfidence → high self-rating but NOT better performance

We can't determine causation from correlation alone.

8.5 When Can We Infer Causation?

Only with **experimental manipulation**:

1. Randomly assign participants to conditions
2. Manipulate one variable
3. Measure the effect on another variable

Because random assignment controls for confounding variables, differences between groups can be attributed to the manipulation.

> ℹ The Memory Experiment
>
> The memory study (abstract vs concrete words) WAS an experiment. Random assignment to conditions allows causal inference: word type **caused** the difference in recall.
>
> The self-rating vs performance study was correlational — no causal inference possible.

9 Part 8: Common Mistakes with Correlation

9.1 Mistake 1: Assuming Linearity

Pearson's r only measures **linear** relationships. Data can have a perfect curvilinear relationship with r = 0.

```
    Y |      •
      |    •   •
      |  •       •
      | •          •
      |•             •
      +-------------
              X
```

This U-shaped pattern has no *linear* correlation, but there's clearly a relationship!

Solution: Always plot your data first.

9.2 Mistake 2: Restricted Range

If you only measure people who are similar on one variable, correlations appear weaker than they really are.

Example: If you only study university students (high ability), you might find no correlation between IQ and grades — but in the general population, there is one.

9.3 Mistake 3: Outliers

A single extreme point can dramatically inflate or deflate r.

```
Without outlier: r = .20      With outlier: r = .75

    •   •                         •   •           •
  •   •                         •   •
    • •                           • •
  •   •                         •   •
```

Solution: Check for outliers; consider Spearman's ρ.

9.4 Mistake 4: Confusing Significance with Strength

With large samples, even tiny correlations become "significant":

- n = 1000, r = .08 → $p < .05$ (significant!)
- But $r^2 = .006$ → explains 0.6% of variance (meaningless!)

Always report and interpret the effect size (r), not just p.

10 Summary

Concept	Key Point
Correlation	Measures relationship between two variables
Scatterplot	Visual display of relationship
Pearson's r	Correlation for linear, continuous data
Spearman's ρ	Correlation for ranked/ordinal data
r = +1	Perfect positive correlation
r = 0	No linear correlation
r = −1	Perfect negative correlation
r^2	Proportion of shared variance
Causation	Cannot be inferred from correlation alone

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11 Correlation Strength Quick Reference

	r		Interpretation
.10	Small	.01	1%
.20	Small-medium	.04	4%
.30	Medium	.09	9%
.40	Medium-large	.16	16%
.50	Large	.25	25%
.70	Very large	.49	49%

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12 Check Your Understanding

Before the lecture, make sure you can answer these questions:

1. What's the difference between a positive and negative correlation?
2. What does r = .40 tell you about the relationship between two variables?
3. When would you use Spearman's ρ instead of Pearson's r?
4. If r = .50, what is r^2 and what does it mean?
5. Why doesn't correlation prove causation? Give an example.
6. Why should you always plot your data before calculating r?
7. In the DataLab, self-rating correlated with performance. What alternative explanations exist besides "confidence causes better performance"?

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13 Further Reading

For additional depth:

- Coolican, H. (2017). Chapter 16: Correlation. VLeBooks
- Vigen, T. (2015). *Spurious Correlations*. tylervigen.com/spurious-correlations — Entertaining examples of correlation without causation.